

Subset-sum landscapes under deformed and random measures: the biased invariant $\Gamma^{(q)}$ and unconditional rigidity

(working title — scaffolding, round 100)

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Abstract

[**TODO** written last.] Papers 1–3 fixed both the sequence and the measure. This paper moves the measure along two axes — bias the coin, and randomise the sequence — and collects the two open questions recorded in paper 3.

1 Introduction

[**TODO** full introduction.] The two deformation axes, and why each is worth taking:

- *Bias.* The classification theorem of paper 1 is a statement about which elements lie on which side of a target; it says nothing about how configurations are weighted. So the whole combinatorial layer survives verbatim when the uniform measure is replaced by the Bernoulli(q) product measure, and the gap series deforms into an explicit one-parameter family. §2 does this, and the answer has a shape worth stating in the introduction: **the fair coin gives the least rugged landscape**.
- *Randomness.* For a random odd sequence the minor-arc input that cost paper 2 fifty rounds is replaced by a second-moment bound. §4 is therefore expected to carry *no ineffective constant at all* — the first such statement in the series. [**TODO** write; see §4.]

2 The biased invariant

Let P_q be the product measure under which each $a_i \in A$ is present independently with probability $q \in (0, 1)$, and write $\text{lm}_q(n)$, $\text{deg}_q(n)$ for the P_q -measures of the sets of strict local minima and of ground states at the target n . The target is taken at the q -tilted centre $n_q = \lfloor qT \rfloor$.

Definition 2.1 (biased gap series).

$$\Gamma^{(q)}(A) = 1 + \sum_{d \geq 1} \left[q^{N_d} + (1 - q)^{N_d} \right].$$

At $q = \frac{1}{2}$ every bracket is $2 \cdot 2^{-N_d}$ and $\Gamma^{(1/2)}$ is the window series W_D of paper 1, hence $\Gamma(A) + (2D + 1)2^{-k}$.

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Remark 2.2 (why the classification survives the deformation). Paper 1’s classification is measure-free: at error d , an overshooting configuration is a strict local minimum exactly when every $a \leq 2d$ is *excluded*, and an undershooting one exactly when every $a \leq 2d$ is *included*. Under P_q those two events have probability $(1-q)^{N_d}$ and q^{N_d} , which is Definition 2.1. Writing $r_B^q(m) = \sum_{U \subseteq B, \sigma(U)=m} q^{|U|}(1-q)^{|B|-|U|}$ and factorising P_q across I_d and B_d ,

$$\text{over}_q = (1-q)^{N_d} r_{B_d}^q(n+d), \quad \text{under}_q = q^{N_d} r_{B_d}^q(n-d-\sigma_d), \quad \deg_q = \sum_{J \subseteq I_d} q^{|J|} (1-q)^{N_d-|J|} r_{B_d}^q(n-\sigma(J)),$$

and the binomial sum over J is exactly 1, so a flat r^q gives the d -th term of Definition 2.1. [\[derived; the assignment verified per stratum, see Remark 2.8\]](#)

2.1 The fair coin is the flattest

Lemma 2.3 (termwise minimality). *For every integer $N \geq 0$ the function $f_N(q) = q^N + (1-q)^N$ on $(0,1)$ is symmetric about $q = \frac{1}{2}$ and convex, and $f_N(q) \geq f_N(\frac{1}{2}) = 2^{1-N}$, with strict inequality for $q \neq \frac{1}{2}$ as soon as $N \geq 2$.*

Proof. $f_N(1-q) = f_N(q)$ is immediate. For $N \geq 2$, $f'_N(q) = N[q^{N-1} - (1-q)^{N-1}]$ and $f''_N(q) = N(N-1)[q^{N-2} + (1-q)^{N-2}] > 0$, so f_N is strictly convex; and $t \mapsto t^{N-1}$ is strictly increasing, so $f'_N < 0$ on $(0, \frac{1}{2})$ and $f'_N > 0$ on $(\frac{1}{2}, 1)$. For $N \in \{0, 1\}$, f_N is the constant 2 or 1. $\square$

Remark 2.4 (formally verified). Lemma 2.3 is machine-checked. `termwise_min` gives the inequality for every N and every $q \in [0, 1]$, `termwise_min_strict` its strictness for $N \geq 2$ and $q \neq \frac{1}{2}$, and `termwise_min_iff` the equality case as a biconditional; the normalisation $2 \cdot (\frac{1}{2})^N = 2^{1-N}$ is `two_mul_half_pow`. The non-strict bound is proved twice, once from convexity of x^N on $[0, \infty)$ and once from Bernoulli’s inequality (`termwise_min'`), so the two proofs check the *statement* against each other and not merely the tactic script. No sorry, and no axioms beyond `propext`, `Classical.choice`, `Quot.sound`. [\[Lean-verified; Pnp/Theory/TermwiseMin.lean\]](#)

Theorem 2.5 (the fair coin minimises the invariant). *For every finite increasing sequence A of odd positive integers with $|A| \geq 2$ and every $q \in (0, 1)$,*

$$\Gamma^{(q)}(A) \geq \Gamma^{(1/2)}(A),$$

with equality if and only if $q = \frac{1}{2}$.

Proof. Sum Lemma 2.3 over d . For strictness it suffices that some $d \leq D$ has $N_d \geq 2$: take any $d \geq (a_2 - 1)/2$, which is at most $D = (a_k - 1)/2$ because $a_2 \leq a_k$. $\square$

Remark 2.6 (what the theorem says). $q(1-q)$ is the per-element variance of the coin. Biasing the coin in either direction lowers that variance and raises the invariant: **the more predictable the coin, the more rugged the landscape**. The effect is exactly symmetric in $q \leftrightarrow 1-q$, as it must be, since complementing every configuration exchanges q with $1-q$ and n with $T-n$.

Corollary 2.7 (closed form for the odd numbers). *For $A = (1, 3, \dots, 2k-1)$ one has $N_d = d$ for $d \leq k$, so in the limit $k \rightarrow \infty$*

$$\Gamma^{(q)} = 1 + \frac{q}{1-q} + \frac{1-q}{q} = \frac{1}{q(1-q)} - 1 \geq 3,$$

with equality only at $q = \frac{1}{2}$: the invariant of the odd numbers is the reciprocal of the coin’s variance, minus one. [\[verified against the summed series to \$5.3 \cdot 10^{-15}\$ over \$q = 0.10, \dots, 0.80\$; e5_r098\]](#)

2.2 Verification

[experimentally confirmed] Measured lm_q/deg_q at n_q against the point prediction of Definition 2.1, relative error:

profile	q	$k = 60$	$k = 100$	$k = 140$
odds	0.20	$1.00 \cdot 10^{-2}$	$3.16 \cdot 10^{-3}$	$1.49 \cdot 10^{-3}$
odds	0.30	$2.63 \cdot 10^{-3}$	$8.02 \cdot 10^{-4}$	$3.75 \cdot 10^{-4}$
odds	0.50	$2.73 \cdot 10^{-4}$	$5.99 \cdot 10^{-5}$	$2.19 \cdot 10^{-5}$
primes	0.30	$2.18 \cdot 10^{-3}$	$5.02 \cdot 10^{-4}$	$2.20 \cdot 10^{-4}$
primes	0.50	$1.73 \cdot 10^{-4}$	$2.26 \cdot 10^{-5}$	$7.32 \cdot 10^{-6}$

There are no free parameters. Two features: the error decays faster than $1/k$ (fitted exponent 2.25 at $q = 0.2$ to 3.0 at $q = 0.5$), and it grows monotonically as q leaves $\frac{1}{2}$, by a factor ≈ 40 from $q = 0.5$ to $q = 0.2$ at fixed k . The second is what the biased flatness hypothesis will have to quantify. **[TODO state the biased (H); design after §2.3.]**

Remark 2.8 (the assignment, and a test that could not decide it). Definition 2.1 is symmetric in $q \leftrightarrow 1 - q$, and so is the exact complementation identity $\text{lm}_q(n) = \text{lm}_{1-q}(T - n)$ — at the q -tilted centre, $T - qT = (1 - q)T$. Neither can therefore distinguish the assignment of q^{N_d} and $(1 - q)^{N_d}$ to the two strata: measured at $k = 100$, the ratio at $q = 0.4$ and at $q = 0.6$ agrees to every printed digit. The discriminating test is per stratum, over q/deg_q against $(1 - q)^{N_d}$ and under q/deg_q against q^{N_d} ; measured, both ratios lie in $[0.9978, 1.0018]$ over $d \leq 6$, whereas a swap would disagree by a factor of 161 at $d = 6$. [experimentally confirmed; e5_r098]

2.3 The bias breaks the reflection symmetry

[experimentally confirmed, shape] Off the q -tilted centre the deviation $\text{dev}_q = (\text{lm}_q/\text{deg}_q)/\Gamma^{(q)} - 1$ behaves differently at $q = \frac{1}{2}$ and away from it, and the reason is structural. Paper 3's transfer function is even in λ because the two strata sit at $\pm\delta_d$ about a common centre — the analytic form of $\text{lm}(n) = \text{lm}(T - n)$. Under bias that symmetry maps q to $1 - q$ rather than to itself, so the odd orders need not cancel, and the first-order coefficient is

$$L^{(q)}(A) = \sum_{d \geq 1} \delta_d [q^{N_d} - (1 - q)^{N_d}],$$

which is antisymmetric in $q \leftrightarrow 1 - q$ and vanishes identically at $q = \frac{1}{2}$. Measured for the odd numbers at $k = 80$: $L^{(q)} = -28.634, -8.843, 0, +8.843, +28.634$ at $q = 0.3, 0.4, 0.5, 0.6, 0.7$, and for the primes $-83.717, -23.677, 0, +23.677, +83.717$.

The data agree. Over the same range of targets, the spread (max over min) of $|\text{dev}_q - \text{dev}_q(0)|$ divided by λ_q and by λ_q^2 :

profile	q	divided by λ_q	divided by λ_q^2
odds	0.30	1.329	9.644
odds	0.50	5.091	1.001
primes	0.30	1.284	8.858

So the response is linear in λ under bias and quadratic at $q = \frac{1}{2}$, where $(\text{dev} - \text{dev}_0)/\lambda^2$ is constant to four digits (19.8186 to 19.8352 across a fivefold range of the target offset). [experimentally confirmed; e6_r100] **[TODO the primes at $q = \frac{1}{2}$ are not a clean test in this**

range: $\text{dev} - \text{dev}_0$ crosses zero inside it, and dropping the single point where it is $2.7 \cdot 10^{-7}$ leaves the quadratic ratio in $[45.06, 48.86]$. Re-measure on a target range that does not straddle the crossing.]

3 The biased transfer function

3.1 The sign of λ , which now matters

Paper 3's transfer function is even in λ , so the sign of the slope is not observable there. Splitting the cosh makes it observable, so we fix it first:

$$\lambda_q(n) = + \frac{d}{dm} \log r_A^q(m) \Big|_{m=n} = \frac{1}{4} \log \frac{r_A^q(n+2)}{r_A^q(n-2)} + O(\cdot),$$

positive below the biased centre. This is the convention under which $\lambda \approx s$; paper 3's def:slope carried the opposite sign and has been corrected, with no numerical consequence there for exactly the reason just given.

3.2 The derivation

Take $P_{q,s}$: element a present with weight $q e^{-sa}$, absent with weight $1-q$, so $p_a = q e^{-sa} / ((1-q) + q e^{-sa})$ and $p_a = q - q(1-q)sa + O(s^2 a^2)$. Two consequences, both absent at $q = \frac{1}{2}$:

- the per-element variance is $a^2 p_a(1-p_a) = q(1-q)a^2(1+O(sa))$, so the layer weight of paper 3's rem:mushift becomes $e^{-\lambda^2 q(1-q)s_d/2}$;
- removing I_d removes mean $q\sigma_d$, so the two strata no longer sit at $\pm\delta_d$ about a common centre. They sit at

$$\delta_d^+ = d + q\sigma_d, \quad -\delta_d^-, \quad \delta_d^- = d + (1-q)\sigma_d.$$

The bias splits the pair asymmetrically, in the ratio $q : (1-q)$.

Hence

$$\Phi^{(q)}(\lambda) = 1 + \sum_{d \geq 1} \left[(1-q)^{N_d} e^{\lambda \delta_d^+} + q^{N_d} e^{-\lambda \delta_d^-} \right] e^{-\lambda^2 q(1-q)s_d/2}. \quad (1)$$

[derived; three internal checks below are exact, the point prediction is verified]

Three checks are built in, and all three hold to machine zero ($0.00 \cdot 10^0$, two profiles, $k = 60$): $\Phi^{(q)}(0) = \Gamma^{(q)}$ of Definition 2.1; the complementation symmetry $\Phi^{(q)}(\lambda) = \Phi^{(1-q)}(-\lambda)$, which holds because $\delta^+(q) = \delta^-(1-q)$; and $\Phi^{(1/2)}$ is paper 3's Φ term by term. [experimentally confirmed; e10_r105]

3.3 Tilted rigidity

Proposition 3.1 (the biased landscape responds at first order). $\Phi^{(q)'}(0) = \sum_{d \geq 1} \left[(1-q)^{N_d} \delta_d^+ - q^{N_d} \delta_d^- \right]$, which vanishes identically at $q = \frac{1}{2}$ and is non-zero otherwise. So rigidity at a biased measure is not flatness: it is a tilt of a definite slope, and the slope is a point prediction with no free parameter.

This is what §2.3 demanded of the derivation, and it is a sharper demand than it looks. The naive guess — keep paper 3's $\delta_d = d + \sigma_d/2$ and merely split the weights — gives instead $\sum [(1-q)^{N_d} - q^{N_d}] \delta_d$, and the two differ by exactly $(\frac{1}{2} - q) \sum_d \sigma_d [q^{N_d} + (1-q)^{N_d}]$: zero at $q = \frac{1}{2}$, and 20–32% of the prediction at the q tested. The measurement therefore decides between them.

3.4 Verification

The estimator must kill constants, because the point prediction of §2.2 carries an $O(10^{-3})$ offset at the biased centre which is the same size as the linear term. So we use a central difference about n_q : with $n_{\pm} = \lfloor (q \pm hq)T \rfloor$,

$$\hat{L}(h) = \frac{(\text{lm}_q/\text{deg}_q)(n_+) - (\text{lm}_q/\text{deg}_q)(n_-)}{\lambda_q(n_+) - \lambda_q(n_-)}.$$

Ratios of $\hat{L}$ to the two candidates, at $h = 0.01$:

profile	q	separation	$\hat{L}/\Phi^{(q)'}(0)$	$\hat{L}/\text{naive}$
odd numbers, $k = 120$	0.30	31.6%	0.9945	0.6805
odd numbers, $k = 120$	0.40	19.9%	0.9976	0.7991
odd numbers, $k = 120$	0.50	—	$\Delta(\text{lm}_q/\text{deg}_q) = 0$ exactly	
odd numbers, $k = 120$	0.60	19.9%	0.9976	0.7991
primes, $k = 120$	0.30	32.3%	0.9981	0.6754
primes, $k = 120$	0.40	20.2%	0.9983	0.7967

Over all 36 discriminating rows (two profiles, $k = 80, 120$, four q , four h), the median $|\hat{L}/\Phi^{(q)'}(0) - 1|$ is 0.0041 against 0.2038 for the naive candidate. The one row above 10% is the primes at $k = 80$, $q = 0.3$, $h = 0.01$, where $\Delta\lambda = 8.5 \cdot 10^{-5}$ and the integer rounding of $n_{\pm}$ is a visible fraction of the step; the same row at $k = 120$ reads 0.9981. At $q = \frac{1}{2}$ the numerator is 0 or $4.4 \cdot 10^{-16}$, below the stated floor — the fair coin has no linear response, which is the content of Proposition 3.1 at its zero. [\[experimentally confirmed; e10_r105\]](#)

The split is also visible stratum by stratum, which tests the offsets directly rather than through their sum: at $k = 80$, $q = 0.3$, $\text{over}_q/\text{deg}_q$ against $(1 - q)^{N_d} e^{\lambda_d^+} e^{-\lambda^2 q(1-q)s_d/2}$ agrees to within 0.24% over $d = 1, \dots, 6$, and $\text{under}_q/\text{deg}_q$ against its partner to within 0.9%. [\[experimentally confirmed; e10_r105\]](#)

3.5 The biased hypothesis

(H_q): $\sum_d [(1 - q)^{N_d} (\delta_d^+)^2 + q^{N_d} (\delta_d^-)^2] < \infty$, uniformly for $q \in [q_0, 1 - q_0]$. Since $\max(q, 1 - q) \leq 1 - q_0 < 1$, the geometric damping of paper 3's (H) survives with base $1 - q_0$, and the constant degrades as $q_0 \rightarrow 0$ — which is the $\sim 40\times$ growth of the §2.2 error already measured between $q = \frac{1}{2}$ and $q = 0.2$. The $\alpha = 1$ threshold of paper 3 is unchanged, the damping argument being base-agnostic. [\[derived; the degradation matches the measured trend\]](#)

4 Random sequences, unconditionally

Before the theorem, the floor it has to stand on. Let A be drawn from the Cramér model on the odd integers: each odd $m \in [3, N]$ is included independently with probability $2/\log m$, and N is fixed by $\mathbb{E}|A| = k$. Nothing is conditioned, so $|A|$ fluctuates from instance to instance. That is deliberate: $kR_A(k)$ is the k -invariant quantity of paper 3's rem:onek, so an ensemble of genuinely different sizes tests the $1/k$ law for free. Twenty instances at each of two sizes, each through the whole paper-3 pipeline with its own N_d and σ_d :

$\mathbb{E} A $	range of k	(H) series	mean kR	sd	sd/mean
120	101–135	133.79–320.79	214.96	10.58	4.92%
180	146–191	132.94–471.77	206.32	6.94	3.37%

at $x = 0.06$; $c_A = kR/100$. Three things, and the third is the one to keep. [\[experimentally confirmed; e7_r102\]](#)

(i) *The hypothesis holds on the ensemble.* All forty instances have a bounded (H) series and a window that tracks the primes'. This is a precondition, not a result: a theorem about random sequences is unconditional only if the random sequences satisfy (H) themselves.

(ii) *The constant concentrates.* The spread across instances is under 5% and falls with k , while the mean of kR moves by 4% between the two sizes — the $1/k$ law of paper 3 surviving an ensemble it was not fitted on.

(iii) *The primes sit inside the ensemble and the odd numbers do not.* Against the $k \approx 180$ ensemble (2.0632 ± 0.0694 in c_A), the primes score $z = -0.39$ and the odd numbers $z = -2.70$, at every one of the four targets. In this statistic the primes are indistinguishable from a random odd sequence of the same density; the odd numbers are a different object.

There is one place where the random model is *not* like the primes, and it is the place that matters for the minor arcs. Write $h_d(q) = \max_{(r,q)=1} (\prod_{a \in B_d} |\cos(\pi r a/q)|)^{1/|B_d|}$. Since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for *every* odd a , the modulus-4 floor $h_d(4) = 1/\sqrt{2}$ is deterministic, not merely universal. Paper 2's vanishing lemma kills the modulus- $2m$ peak (m odd) as soon as B_d contains an element $\equiv 3 \pmod{6}$. A set of primes contains exactly one such element, namely 3, and 3 leaves B_d at $d \geq 2$, so the primes keep $h_d(6) = \sqrt{3}/2$; a random odd sequence contains about $|B_d|/3$ of them and loses the peak at every d . Measured over $q \leq 120$ at $d = 1, 2, 5$: every random instance has $\max_{q \geq 3} h_d(q) = 1/\sqrt{2}$ attained at $q = 4$, and the primes have $\sqrt{3}/2$ at $q = 6$. [\[experimentally confirmed; e7_r102\]](#)

So the random theorem should come out with the geometric rate $1/\sqrt{2} = 0.7071$ per element rather than the $\sqrt{3}/2 = 0.8660$ of papers 1–2: *the primes are the harder case, and the modulus-6 ripple is a structural feature of the primes that the random model destroys.*

[TODO fable-5: for random odd A the minor-arc input is a second-moment bound $\mathbb{E}|S_A|^2 = b$ plus Borel–Cantelli on a θ -net — no Kumchev, no Siegel–Walfisz, no Siegel constant. If it lands it is the trilogy's first zero-ineffective-constant rigidity theorem. The measurements above are the floor it must clear, and they sharpen the brief: the rate to aim for is $1/\sqrt{2}$, not $\sqrt{3}/2$.]

5 Two questions inherited from paper 3

(i) *The c_d fold-in is settled, and it goes paper 3's way.* The share of the residual that c_d carries falls like $k^{-2\alpha}$ — measured exponents 1.98, 2.35, 2.99 against predicted 2.00, ≈ 2.4 , 3.00 on the odd numbers, the primes and $\alpha = \frac{3}{2}$, with the squares already below the numerical floor. So Φ as printed is right for every larger k and c_d belongs to E_k permanently; paper 3's rem:cd now says so. [\[experimentally confirmed, four profiles, \$k \leq 300\$; e8_r103\]](#)

(ii) $c_A = c'_A$ is still open, but sharper. Carried to $k = 300$ the ratio reaches 1.0064 (odd numbers), 1.0001 (primes), 0.9984 ($\alpha = \frac{3}{2}$), and the gap closes like $k^{-2.4}$ — faster than the $1/k$ law both constants sit on. [\[experimentally confirmed; e8_r103\]](#) **[TODO** the mechanism is still missing, and a measured rate of convergence is evidence about a limit, not a proof of one. F45 stands: substituting s for λ makes the residual three times worse.]

6 Outlook

[TODO the algorithmic reading — local search on structured instances — which is where this programme started. Outlook only; no claims.]

7 Honest scope

- *Lean-verified*: Lemma 2.3 in all three forms (Remark 2.4), including the equality case.
- *Proved*: Lemma 2.3, Theorem 2.5, Corollary 2.7; and Remark 2.2's derivation of Definition 2.1 modulo the biased flatness hypothesis, which is not yet stated.
- *Experimentally confirmed, with the range stated*: the point prediction (§2.2, two profiles, five q , three k), the stratum assignment (Remark 2.8), the linear response under bias (§2.3).
- *Derived and verified*: $\Phi^{(q)}$ of (1) and Proposition 3.1 — three exact internal checks, and the first-order point prediction confirmed against the naive alternative on 36 discriminating rows (§3.4); the random-ensemble floor of §4; the c_d verdict of §5.
- *Not yet written*: §§6, the theorem of §4, and the passage from (H_q) to a rigidity statement.

References

- [1] K. Amauchi, *The gap series of an integer sequence*. Manuscript, 2026.
- [2] K. Amauchi, *Asymptotic flatness of subset-sum landscapes of primes*. Manuscript, 2026.
- [3] K. Amauchi, *The transfer function of subset-sum landscapes*. Manuscript, 2026.